No-go theorem for quantum realization of extremal correlations
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Sujan V.K.
Indian Institute of Science Education and Research, Pune 411008, India

Ravi Kunjwal
Université libre de Bruxelles, QuIC, Brussels, Belgium
Aix-Marseille University, CNRS, LIS, Marseille, France

We consider quantum realizations of extremal correlations in any experimental set-up (including all Bell scenarios) where an arbitrary finite set of measurements is implemented on a given preparation. Such an experimental set-up is described by a *contextuality scenario* [1].

We prove that, for all contextuality scenarios, no extremal and non-deterministic correlation can be achieved using projective quantum measurements, *i.e.*, there exists no quantum state and no set of projective measurements, for any contextuality scenario, that can achieve such correlations (a special case of this is the impossibility of Popescu-Rohrlich or PR boxes in quantum theory [8]).

The above no-go result, in turn, follows as a corollary of a more general no-go theorem that holds when general quantum measurements (or POVMs) are taken into account. This general no-go theorem entails that no non-trivial quantum realization of an extremal and non-deterministic correlation exists, *i.e.*, any "quantum" realization must be simulable by discarding the quantum system at hand and using a classical source of randomness.

Definitions

We briefly recall some preliminary concepts that we use to prove our results. The key concepts we need are: contextuality scenarios, probabilistic models, quantum models, and quantum realizations of probabilistic models. All Bell scenarios are contextuality scenarios, but not conversely.

Definition 1 (Contextuality scenario)**.** *A* contextuality scenario *is a hypergraph H with set of vertices $V(H)$ and set of edges $E(H) \subseteq 2^{V(H)}$ such that*

$$\bigcup_{e\in E(H)} e = V(H). \tag{1}$$

Definition 2 (Probabilistic model)**.** *Let H be a contextuality scenario. A* probabilistic model *on H is an assignment $p : V(H) \to [0,1]$ of a probability $p(v)$ to each vertex $v \in V(H)$ such that*

$$\sum_{v\in e} p(v) = 1 \quad \forall e \in E(H). \tag{2}$$

We denote the set of (general) probabilistic models on H by $\mathcal{G}(H)$. This set defines a convex polytope and we refer to the vertices of this polytope as extremal probabilistic models*.*

Definition 3 (Quantum models)**.** *Given a contextuality scenario H, $p \in \mathcal{G}(H)$ is a* quantum model *if there exists a Hilbert space $\mathcal{H}$, a quantum state $\rho \in \mathcal{B}(\mathcal{H})$, and a projection operator $P_v \in \mathcal{B}(\mathcal{H})$*

associated to every $v \in V(H)$, such that every hyperedge $e \in E(H)$ forms a projective measurement, i.e.,

$$\sum_{v\in e} P_v = \mathbb{1} \quad \forall e \in E(H), \tag{3}$$

and this construction reproduces the probabilistic model p, i.e.,

$$p(v) = \operatorname{tr}(\rho P_v) \quad \forall v \in V(H). \tag{4}$$

We denote the set of quantum models on H by $\mathcal{Q}(H)$, where $\mathcal{Q}(H) \subseteq \mathcal{G}(H)$.

We now define the notion of quantum realization of a probabilistic model (which may not be a quantum model in the sense of **Definition** 3).

Definition 4 (Quantum realization of a probabilistic model)**.** *Given a contextuality scenario H, a* quantum realization *of $p \in \mathcal{G}(H)$ is given by a Hilbert space $\mathcal{H}$, a quantum state $\rho \in \mathcal{B}(\mathcal{H})$, and a set of positive semidefinite operators $\{E_v \in \mathcal{B}(\mathcal{H})\}_{v\in V(H)}$, i.e.,*

$$(\mathcal{H}, \rho, \{E_v\}_{v\in V(H)}), \tag{5}$$

such that

$$\sum_{v\in e} E_v = \mathbb{1} \quad \forall\, e \in E, \text{ and} \tag{6}$$

$$p(v) = \operatorname{tr}(\rho\, E_v) \quad \forall v \in V(H). \tag{7}$$

Clearly, every quantum model (by definition) admits a quantum realization where the positive operators are projectors. However, the converse is not true, *i.e.*, there exist probabilistic models $p \in \mathcal{G}(H) \backslash \mathcal{Q}(H)$ that admit a quantum realization. In fact, *all* probabilistic models $p \in \mathcal{G}(H)$ admit the following trivial quantum realization:

$$(\mathcal{H}, \rho, \{p(v)\mathbb{1}\}_{v\in V(H)}), \tag{8}$$

for arbitrary choices of $\mathcal{H}$ (with dimension ≥ 1) and ρ. However, there also exist non-trivial quantum realizations of probabilistic models that are not in $\mathcal{Q}(H)$, *e.g.*, in the case of Specker's scenario [9, 10, 11, 12], where one can come up with non-trivial quantum realizations that exhibit quantum correlations stronger than what is possible with projective measurements, such as the violation of the Liang-Spekkens-Wiseman (LSW) inequality [13, 14, 15, 16]. Below we formalize the notion of a trivial quantum realization:

Definition 5 (Trivial quantum realization of a probabilistic model)**.** *Given a contextuality scenario H, a* quantum realization *$(\mathcal{H}, \rho, \{E_v\}_{v\in V(H)})$ of $p \in \mathcal{G}(H)$ is said to be trivial if it takes the following general form:*

$$\rho := \Lambda_r \oplus 0_{d-r}, \tag{9}$$

$$E_v := p(v)\mathbb{1}_r \oplus E_v^{(d-r)} \quad \forall v \in V(H), \tag{10}$$

where r is the rank of ρ, d is the dimension of $\mathcal{H}$, and in the eigenbasis of ρ we have $\Lambda_r := \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_r)$, $0_{d-r} := \operatorname{diag}(0, 0, \ldots, 0)$, $\mathbb{1}_r := \operatorname{diag}(1, 1, \ldots, 1)$, and $E_v^{(d-r)}$ are positive operators subject only to the normalization constraints $\sum_{v\in e} E_v^{(d-r)} = \mathbb{1}_{d-r}$ for all $e \in E(H)$.

It should be clear why we call such a quantum realization "trivial": any probabilistic model $p \in \mathcal{G}(H)$ that can be reproduced in the above manner on a Hilbert space of dimension d can also be reproduced by the following quantum realization on a Hilbert space of dimension $1 \leq r \leq d$:

$$(\mathbb{C}^r, \Lambda_r, \{p(v)\mathbb{1}_r\}_{v\in V(H)}), \tag{11}$$

which is equivalent to just discarding the quantum system at hand and generating the probabilistic model using a classical source of randomness.

Results

We obtain the following results:

- Our main no-go theorem, *i.e.*,

  Theorem 1 (No-go theorem for non-trivial quantum realizations)**.** *For any contextuality scenario, quantum realizations of its extremal probabilistic models must necessarily be trivial.*

- A first corollary of our no-go theorem, *i.e.*,

  Corollary 1 (No-go for projective quantum realizations)**.** *For any contextuality scenario, the set of quantum models excludes indeterministic extremal probabilistic models, i.e., it is impossible to realize an extremal probabilistic model that is indeterministic using projective quantum measurements.*

- A second corollary of our no-go theorem, recovering the central claim of Ref. [8] in a simpler and more direct manner than the original proof:

  Corollary 2 (No-go for quantum realizations in Bell scenarios)**.** *No Bell-nonlocal vertex of the nonsignaling polytope in any Bell scenario admits a quantum realization.*

We also discuss three key aspects of our results:

- *Naimark dilations, i.e.*, how they differ when it comes to general contextuality scenarios vis-à-vis Bell scenarios: they preserve nonsignaling relations in the latter case but fail to preserve the corresponding nondisturbance relations in the former case, thus posing a conceptual hurdle for the "Church of the Larger Hilbert space" [17].
- *Beyond contextuality scenarios arising from PVMs, i.e.,* how we need to consider contextuality scenarios that arise from POVMs [13, 14, 15] rather than only the standard ones realizable with PVMs.
- *From no-go theorems for extremal determinism to a no-go theorem for extremal indeterminism, i.e.,* how our results can be seen as counterpart to Bell, Kochen-Specker, and Gleason's theorems, just from the opposite end, *i.e.*, extremal indeterminism rather than extremal determinism.

We have uncovered a structural constraint on the intrinsic indeterminism of quantum theory by proving that such indeterminism cannot be extremal for any set of quantum measurements.[1] It seems to be a general feature of quantum theory that extremal correlations consistent with some general physical principle that are achievable by quantum theory are also achievable in its classical deterministic limit, whether it is in the case of Bell scenarios [8] (the principle here being nonsignaling), in contextuality scenarios (as we show here, the principle here being nondisturbance or consistency of marginals [18, 19, 20]), or even in scenarios with indefinite causal order [21] where standard quantum theory is extended to process matrices [22] with process functions [23, 24] as their classical deterministic limit [21, 25, 26] (the principle here being logical consistency [23]). Hence, extremal correlations that are impossible in a classical deterministic theory in each of these cases are also impossible in quantum theory, *i.e.*, they are post-quantum correlations (PR-boxes being perhaps the most well-known example).

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